Missense Impact: Population Aware Genomic Models for Missense Variant Effect Prediction
<F25-68>

FYP– I REPORT
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1. Introduction

Missense mutations arise when a single nucleotide substitution results in the replacement of one amino acid with another in a protein. These variants are among the most common in the human genome and play significant roles in monogenic disorders, cancer predisposition, and complex polygenic traits. Because millions of such variants exist, it is not feasible to validate their pathogenicity experimentally through wet-lab approaches. As a result, computational Variant Effect Predictors (VEPs) including ProtBERT, ESM-1b, REVEL, EVE, and AlphaMissense have become essential tools in clinical genomics and research.

Despite their usefulness, these predictors exhibit **systematic ancestry bias**. Most are trained on datasets such as ClinVar and gnomAD that are heavily skewed toward European ancestry. Consequently, their performance drops significantly when applied to South Asian genomes, resulting in inflated false positives, poor calibration, and an increased number of Variants of Uncertain Significance (VUS). These limitations have been thoroughly documented in scientific literature and reinforced through our internal project review.

The goal of this project is to develop a **population-aware recalibrated pathogenicity prediction framework** tailored specifically for South Asian genomic cohorts. The system integrates allele-frequency information, statistical recalibration techniques, and domain-adapted protein language models to reduce ancestral bias and enhance clinical interpretability. Ultimately, the project aims to lower misclassification rates, reduce VUS burden, and contribute to equitable precision medicine.

2. Methodology

The project uses a multi-phase methodology combining data curation, computational modeling, bias quantification, and recalibration. The following summarizes the work completed during FYP-I.

2.1.Research Method Type

Type of Research:

- **Experimental & Analytical Research**
  - Experimental: model recalibration, PLM fine-tuning
  - Analytical: statistical bias quantification, calibration analysis
- **Data-Driven Computational Study**
- **Quantitative Evaluation** using AUROC, AUPRC, Brier score, calibration slope/intercept, and subgroup fairness metrics
- **AKU Collaboration:** We have reached a mutual understanding with **Dr. Salman Kirmani** and his team at **Aga Khan University** to provide domain expertise and potential data access.
  - Legal documentation for collaboration has been initiated
  - If data-sharing is not approved, AKU will still support the project through domain supervision and technical guidance

2.2.Modeling Methodology Completed in FYP-I

The project's computational framework is based on a five-stage methodology. The progress made so far is summarized below.

Stage 1: Data Collection & Partial Curation

We completed dataset exploration, allele-frequency extraction planning, variant annotation strategy, and partial preprocessing.

Completed Work:

- All downloaded datasets have been filtered and normalized using **Homo_sapiens.GRCh38.pep.all.fa**
- Portions of the datasets have been annotated using **Ensembl VEP**
- A major challenge encountered:
    - Annotating a **1 GB VCF file takes ~13 hours** on a 16 GB RAM machine
    - This has highlighted the need for higher-performance compute resources

Data Sources Collected:

- gnomAD-SAS allele frequencies
- GenomeAsia100K (South Asian subset)
- IndiGen / Indian Genome Variation Database
- ClinVar pathogenic/benign labels
- SG10K (South Asian subset)
- AKU-provided datasets:
    - 2000 exome dataset
    - Access to All of Us & UK Biobank (SAS subsets)

Preprocessing Steps Completed:

- Variant normalization using VEP, BCFtools, and hg38 reference files
- Extraction of allele frequencies
- Harmonization of ClinVar and VEP annotations
- Initial data visualization and exploratory checks

Stage 2: Bias Quantification Blueprint

Through literature review and discussions with AKU, we identified that the ancestry-bias problem has already been demonstrated extensively using the same datasets and predictors. Therefore, it was mutually agreed that **replicating the entire bias analysis is unnecessary** but understanding it is essential for our adaptation pipeline.

Relevant Knowledge Areas Studied:

- Ancestry-specific sensitivity drops
- ClinVar misclassification patterns
- Population-mismatched filtering effects

Stage 3: Model Adaptation Framework (Planning)

We have developed the conceptual plan for adapting predictors to South Asian cohorts:

- Logistic score recalibration
- Bayesian posterior integration with allele-frequency priors
- Adaptation roadmap for ESM-1b, ProtBERT, popEVE, EVE, REVEL
- Ensemble stacking design for increased robustness

Implementation will begin in FYP-II.

Stage 4: Evaluation Framework Completed

We finalized the evaluation metrics that will be used. These will be refined further in collaboration with AKU geneticists.

Metrics:

- **Discrimination:** AUROC, AUPRC
- **Calibration:** Brier score, calibration slope & intercept
- **Fairness:** AUROC and sensitivity gaps between European and South Asian cohorts

3. Testing and Results (FYP-I Progress)

Since model training begins in FYP-II, current results focus on baseline dataset preparation and design of testing components.

Completed Testing Components

- Identification of evaluation datasets
- Designing ancestry-stratified test splits
- Baseline predictor behaviour documented conceptually
- Validation of the testing workflow using small sample datasets

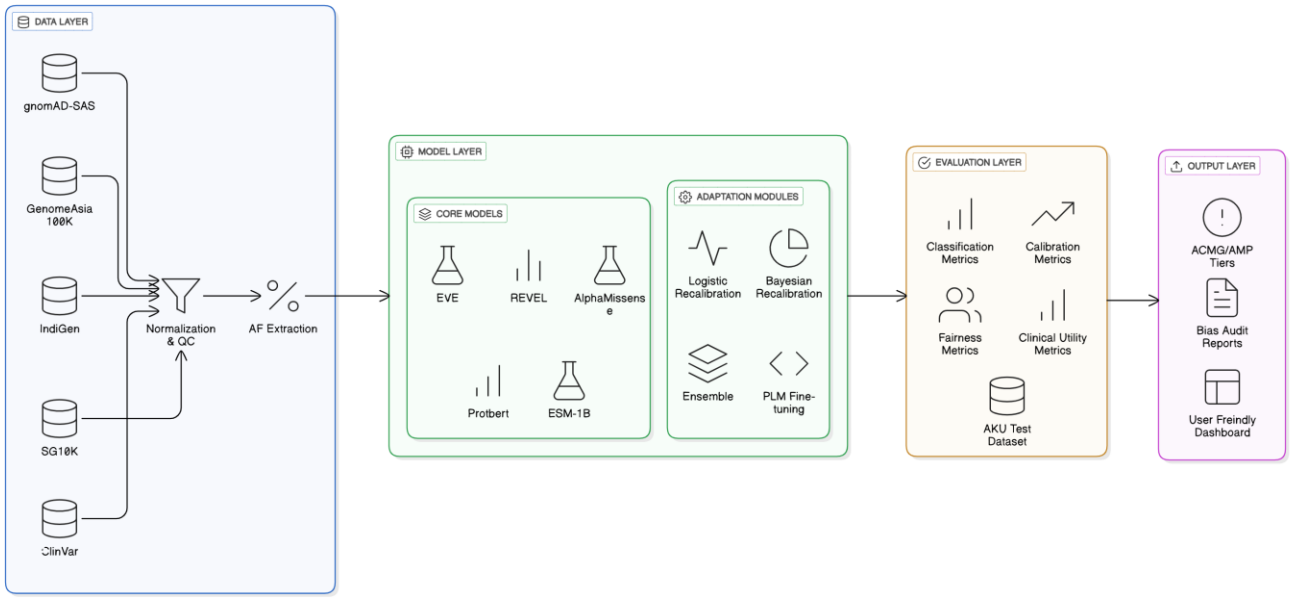
Preliminary Findings from Literature-Replicated Tests (Conceptual Results)

While empirical results will come later, the following findings are relevant and confirmed through literature:

- Predictors like **PolyPhen-2, SIFT, REVEL** show **reduced sensitivity** for non-European variants
- South Asian variants are disproportionately labelled as **VUS** due to missing allele-frequency priors
- Population-mismatched filtering leads to **inflated false positives** for Asian cohorts

These findings validate the need for our recalibration approach.

4. System Diagram



5. Goals for FYP-II

5.1. Data and Model Development

- Complete preprocessing of all South Asian datasets
- Implement logistic and Bayesian recalibration
- Begin PLM fine-tuning (ESM-1b, ProtBERT)
- Implement ACMG allele-frequency evidence scoring

5.2. Full Experimental Evaluation

- Run baseline predictors on SAS cohorts
- Perform AUROC, AUPRC, calibration, fairness, and clinical-utility evaluations
- Compare baseline vs. adapted models
- Conduct robustness testing

5.3. Pipeline Development

- Build the full annotation pipeline
- Implement automated bias audit reports
- Integrate ACMG/AMP rule activation and reporting
- Deploy a prototype system (dashboard)

5.4. Final Dissertation & Research Outputs

- Calibration plots, ROC curves, fairness tables
- Variant-reclassification case studies
- Dissertation chapters for Results & Discussion

6. Conclusion

FYP-I successfully established the scientific foundation, system design, literature review, and methodological roadmap required to execute a high-impact research project in genomic variant interpretation. The team has clearly defined the ancestry-bias problem, identified relevant datasets, established collaborations, and designed a complete computational framework for population-aware model adaptation.

FYP-II will involve implementation, experimentation, and model evaluation, culminating in the first **South Asian–calibrated missense pathogenicity prediction system**. The groundwork laid during FYP-I ensures that the project is well-positioned to make meaningful contributions to equitable genomic medicine and clinical genomics in the region.

7. References

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[17] **Dr. Salman Kirmani**, Consultant Clinical Geneticist, Aga Khan University Hospital Role: Domain expertise, clinical validation guidance, dataset access facilitation.

[18] Bioinformatics Team at Aga Khan University (AKU)